

SAMARPAN MANI GAUTAM

Chemical Engineering Student

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PROFILE

Final-year Chemical Engineering student at Kathmandu University with a dedicated interest in sustainable energy, process optimization, and data-driven environmental modeling. Experienced in conducting techno-enviro-economic assessments, hybrid mechanistic-ML modeling, and system fabrication. Committed to applying computational modeling, machine learning, and automation to solve environmental and process-engineering challenges.

EDUCATION

Kathmandu University Fourth Year Student, Department of Chemical Engineering Expected Graduation: 2026	2022 – Present
Gandaki Boarding School Secondary Education Examination (SEE) - Grade 10	2019
Gandaki Boarding School School Leaving Certificate (+2) - Grade 12	2022

RESEARCH INTERESTS

- **Sustainable Process Engineering:** Decarbonization pathways for heavy industries (Cement & Hydropower integration).
- **Reactor Design & Advanced Oxidation Processes:** Design and fabrication of Cyclonic Plasma Reactors for wastewater treatment.
- **Environmental Data Science & Remote Sensing:** Satellite-ML frameworks (MODIS, TROPOMI, ERA5) for air quality estimation and monitoring-network gap analysis.
- **Nanotechnology:** Synthesis of TiO_2 from biomass (banana ash) for targeted drug delivery systems.
- **Computational Modeling:** CFD (ANSYS Fluent) and machine-learning surrogates for multiphase and environmental systems.
- **Smart Preservation:** IoT-integrated sensor systems for post-harvest loss reduction (Cold Storage).

PUBLICATIONS & PATENTS

Peer-Reviewed Journal Article

- **Published**
 - Shrija Mandal, Ashal Adhikari, Aashish Chaulagain, Ananta Thapa, **Samarpan Mani Gautam**, Sunil Prasad Lohani, Bibek Uprety, "Techno-enviro-economic assessment of hydropower-driven decarbonization pathways for Nepalese cement industry," *Journal of Environmental Chemical Engineering*, 2024, 12(6), 114729. [Q1, IF: 7.7] <https://doi.org/10.1016/j.jece.2024.114729>
- **Under Review**
 - **Samarpan Mani Gautam**, Jigyasu Ghimire, "Synthesizability in AI-Driven Molecular Generation: The Gap, the Metrics, and Strategies for Closing the Distance Between Computation and the Laboratory Bench," *Journal of Computer-Aided Molecular Design, Springer Nature*, under review.

- Udhyan Shah, **Samarpan Mani Gautam**, Rabi Gupta, Ramesh Kumar Maskey, Kushal Shrestha, Jigyasu Ghimire, "Experimental Investigation of Valve Closure Impacts on Surge Tank Transients: A Study at WEIR Laboratory," *IOP Conference Series*, under review.
- **Submitted**
 - **Samarpan Mani Gautam**, Jigyasu Ghimire, Dilip Kumar Rajak, Santosh Kumar Rai, "Gas Holdup Analysis and Prediction in Bubble Columns Using a Hybrid Non-Equilibrium Drift-Flux and Machine Learning Approach," *Journal of Non-Equilibrium Thermodynamics*, De Gruyter, under review.

Patents

- **Cyclonic Plasma Reactor Housing** – Indian Patent Application No. 501052-001, Filed: 2026-05-06, Accepted: 2026-07-08.

CONFERENCES & POSTERS

- **Samarpan Mani Gautam**, Udhyan Shah, Suresh Acharya, Dilip Kumar Rajak, Jigyasu Ghimire, Liza Dev Pradhan Shrestha, "High-Resolution PM_{2.5} Estimation for Nepal: A Two-Stage Machine Learning Framework: Satellite Remote Sensing, Ground-Station Scarcity & Regulatory Reform," Poster presented at **SPACECON Nepal**, Kathmandu, Nepal. <https://spacecon.antarikhya.org.np/>

WORK EXPERIENCE & LEADERSHIP

WEIR Synergy (Water, Energy & Infrastructure Research)

2025 – Present

Vice-President | Researcher in Cyclonic Plasma Reactor

- Leading the **design and fabrication** of a Cyclonic Plasma Reactor for advanced water treatment.
- Utilizing **ANSYS Fluent** for CFD modeling of tangential flow dynamics and vortex stability.
- Conducting **experimental validation** of plasma generation and pollutant degradation efficiency.

IOT & Robotics Trainer

2025

Siddhartha College and Secondary School, Banepa-8, Nepal

- Delivered a comprehensive 3-month curriculum covering the fundamentals of **Embedded Systems**.
- Facilitated hands-on workshops on **Arduino programming** and basic robotics assembly.
- Mentored students in building real-world projects, including automated light systems.

RESEARCH GRANTS & ACADEMIC PROJECTS

Research Grants

- **Nepal Academy of Science and Technology (NAST): Design and Fabrication of Scaled-Up Cold Storage Model with Sensor-Based Regulation**

2024 – 2025

Grant Amount: NPR 1,000,000 (10 Lakhs) | **Status:** Contract Signed (Funding Pending)

- Leading the engineering design for a smart cold storage system to reduce post-harvest losses.

Academic Projects

- **Fabrication of Scaled-Up Cold Storage Model with Integration of Sensor Based Regulation**
 - Developed a physical prototype and conducted thermal stability simulations to optimize insulation.
 - Demonstrated hardware-software integration for food preservation applications.
- **Automated Humidity Control System Using Humidifier and Dehumidifier with Sensor-Based Regulation**
 - Engineered an integrated system using DHT sensors and microcontrollers for precise control.
 - Designed logic for alternating activation based on real-time environmental feedback.
- **Prospects of Utilizing Titanium Dioxide (TiO₂) Extracted from Calcined Banana Ash in Drug Delivery**
 - Investigated the synthesis of TiO₂ from banana ash via calcination for use as a drug carrier.
 - Analyzed biocompatibility and chemical transformation pathways from biomass minerals.

2022

2024

2025

TECHNICAL SKILLS

- **Process Simulation & Design:** Aspen Plus, ANSYS Fluent (CFD), AutoCAD, P&ID, Reactor Design, Plant Design.
- **Geospatial & CAD:** QGIS, FreeCAD, AutoCAD, Fusion 360, 3D Modeling, Prototype Fabrication.
- **Programming & Data Science:** Python (Pandas, NumPy, Scikit-learn), C/C++, $\text{\LaTeX}$.
- **Automation & IoT:** Arduino, Sensor Integration (DHT, Pressure, Flow), Microcontrollers, Circuit Design.
- **Soft Skills:** Technical Writing, Project Management, STEM Mentorship, Research Methodology.

AWARDS AND HONORS

Aspen Technology Global Academic Innovators Challenge – Country Winner (Nepal)

2025

- Recognized as the Country Winner for achieving the highest score in Nepal, representing Kathmandu University.
- Selected among top submissions for technical quality, innovation, and application of simulation tools.